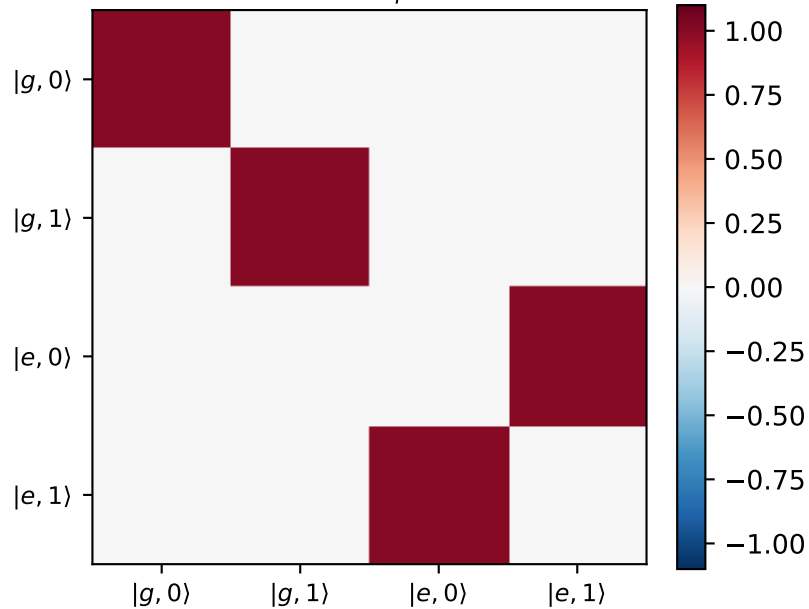
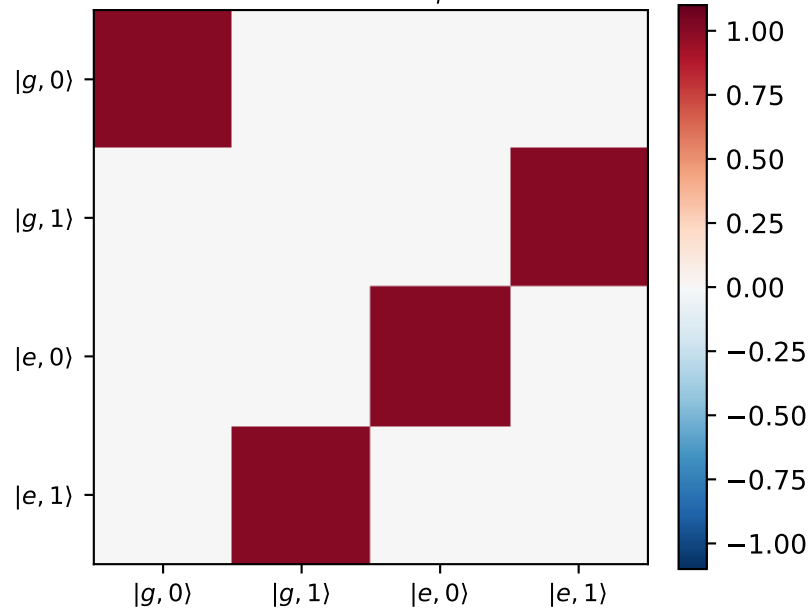


$$\text{Logical sub-gates: } U_{\text{target}} = (I_q \otimes H_c) \cdot \text{CNOT}_{c \rightarrow q} \cdot \text{CNOT}_{q \rightarrow c}$$

 $\text{CNOT}_{q \rightarrow c}$

 $\text{CNOT}_{c \rightarrow q}$

 $I_q \otimes H_c$
